ParkFinder: A Simple Parking Alert Program

Final Project Proposal

Name	Email	Stevens ID
Jennifer Salto	jsalto@stevens.edu	20019356
Reza Jangi	rjangi@stevens.edu	20020365

Section 1: Project Description

1.1 The Problem

Locating a parking space in Hoboken is a constant challenge for residents, commuters, and visitors. The city accommodates approximately 60,000 residents within around one square mile, resulting in significantly less curb space than available vehicles. A resident permit is affordable (approximately $95 annually for an initial vehicle in mid-2026) but merely allows for competing for street spaces, without ensuring a specific spot, causing numerous cars to look for the limited openings simultaneously. Drivers frequently go around block after block without any indication of an available space until they spot it, leading to wasted time and fuel, increased congestion, and heightened stress levels. Drivers who abandon their vehicles on the street are directed to garages that charge about $175–$325 or more each month, and a misstep on the street can result in a ticket, a boot, or having the car towed. The main issue is that drivers lack real-time data on the actual location of available parking spots.

1.2 The Solution

We present ParkFinder, a straightforward Python application that notifies a driver when a parking space opens up. The design presumes that every parking space is equipped with a sensor that indicates if a vehicle is present. The application initializes the initial condition of each location, processes the sensor updates individually, and identifies when a location transitions from occupied to unoccupied. When that occurs, it generates an alert specifying the location and records the alert in a log file. Following the most recent reading, it indicates the number of available spots.

Scope: The installation of physical sensors falls beyond the course project's scope; therefore, we emulate the sensor data using a brief file containing timestamped readings from five locations on Washington Street in Hoboken, as clearly indicated in the README. The program's logic (loading locations, updating conditions, identifying changes, and sending notifications) would function identically with an actual sensor input. The sample data does not reflect actual parking availability, and ParkFinder is not a substitute for checking the posted signs.

1.3 Data Information

The program uses two input files and produces one output. parking_spots.csv has one row per sensor-equipped spot and gives its starting status. sensor_readings.csv holds timestamped occupied/not-occupied readings that the program reads one at a time to mimic a live feed. For output, alerts_log.csv records every alert that was sent. The five sample spots and seven sample readings are simulated for the class demonstration.

parking_spots.csv field	Type	Description
spot_id	str	Unique ID of the sensor/spot (101–105)
location	str	Where the spot is (for example, Washington St near 2nd Street)
occupied	int (1/0)	Starting status: 1 = occupied, 0 = free

sensor_readings.csv field	Type	Example	Meaning
timestamp	str	2026-11-02 08:05:00	When the sensor reported
spot_id	str	101	Which spot the reading is for
occupied	int (1/0)	0	New status; 1 → 0 means the spot just became free (alert)

alerts_log.csv field	Type	Description
timestamp, spot_id	str	When the spot became free and which spot it is
location, message	str	Where, plus alert text, e.g., Parking available at Washington St near 2nd Street (spot 101)

Section 2: Plan to Meet Requirements (Part 1)

1. **Functions:** load_spots() reads the starting spot data into a dictionary, and run_tracker() reads the sensor updates, detects changes, and writes alerts. The program is a single module, parkfinder.py.
2. **Data I/O:** reads parking_spots.csv and sensor_readings.csv with the csv module and a pathlib file path; writes alerts_log.csv.
3. **Control flow:** a for loop over the sensor readings; if statements to skip unknown spot IDs and to alert only when a spot changes from occupied to free.
4. **Types:** mutable – dict (spot_id to its location and status) and list; immutable – str and bool.
5. **Input validation:** a reading for a spot_id that is not in the spot file is skipped with a message instead of crashing the program.
6. **Documentation:** a module docstring, a docstring for each function, and explanatory comments.
7. **__name__:** parkfinder.py ends with an if __name__ == "__main__": block that runs the tracker.

Section 3: Plan to Meet Requirements (Part 2)

We will list the following component in README.md:

1. **List comprehension:** collects the IDs of the spots that are free after the last reading, which gives the final free-spot count.

Section 4: Timeline

When	Milestone
Week 1	Finalize scope; create the sample spot and sensor data; create the public GitHub repo and README
Week 2	Write load_spots() and run_tracker() with alert detection and the alerts_log.csv output
Week 3	Test with the sample readings and edge cases (unknown spot ID, no change in status)
Week 4	Record the 3–4 minute video (pitch + live demo) and upload to YouTube as Unlisted
Week 5	Finish README (run steps, requirement mapping, contributions); check at least five commits per member; test on a clean environment; submit repo link by Dec 16

Team roles: Member 1 – program logic (loading spots, detecting changes, alerts); Member 2 – sample data, testing, README, and video. All members commit to GitHub and appear in the video.

Sources

SpotAngels, "The Ultimate Guide to Hoboken Street Parking" (Feb. 2025). https://www.spotangels.com/blog/ultimate-guide-hoboken-street-parking/

Prodigy Real Estate, "Parking in Hoboken, NJ: A Buyer's Guide" (June 2026). https://prodigyre.com/blog/parking-in-hoboken-nj-a-buyers-guide

Das, Siddharth. “A Novel Parking Management System, for Smart Cities, to Save Fuel, Time, and Money.” 2019 9th Annual Computing and Communication Workshop and Conference (CCWC), IEEE, 2019, https://doi.org/10.1109/CCWC.2019.8666537

City of Hoboken. “Parking FAQs.” City of Hoboken, https://www.hobokennj.gov/resources/parking-faqs. Accessed 25 Oct. 2026.

Hoboken Historical Museum. “Hoboken, New Jersey: The Mile Square City—A Brief History.” Hoboken Historical Museum, https://hobokenmuseum.org/unused_content/hoboken-new-jersey-the-mile-square-city-a-brief-history/. Accessed 25 Oct. 2026.

Intact Public Entities. Risk Management Considerations for Parking Lot Safety. Intact Public Entities, May 2020, https://www.intactpublicentities.ca/storage/media/files/COE/Risk_Management_Considerations/Risk_Management_Considerations_for_Parking_Lot_Safety_0520.pdf